56 PERCENT OF ALL INTERNET TRAFFIC IS FROM AN AUTOMATED SOURCE, SUCH AS HACKING TOOLS, SCRAPERS AND SPAMMERS, IMPERSONATORS, AND BOTS

Data passes from towers to mobile switching center

SWITCHING CENTER

INTERNET SERVICE PROVIDER

Through an ISP, the cell phone is connected to wider Internet

Core routers direct extremely high volumes of data in the Internet backbone

CORE ROUTER

INTERNET BACKBONE

ISP connects to cable or phone companies who control the lines

Internet's major data routes are known as its backbone

Undersea fiber-optic cables carry Internet traffic across seas and oceans, linking continents

TELEPHONE EXCHANGE

INTERNET SERVICE PROVIDER

DATA CENTER

Data is routed along fiber-optic cables or telephone lines

Through an ISP, computers can access the Internet

Data centers contain large systems of computers to handle vast amounts of information

Internet backbone
The principal routes that Internet traffic passes through are known as the Internet backbone. These paths connect major networks and core routers. In order to cope with the extremely high volumes of data passing through every second, much of the backbone is composed of large bundles of fiber-optic cables.

Reassembled image is checked for errors

Error-free image shown at receiver's end

4 Data received
The packets are reassembled, and the message is checked for any errors—for example, to make sure no packets are missing or corrupted.

Packets travel independently by different routes

3 Routing packets
Each packet is routed through the Internet infrastructure, often through different paths. Using multiple paths ensures that the entire message is not lost if a link is broken.

Internet connects computers via hub points called routers

Packets put back together in right order

CAN YOU BREAK THE INTERNET?

Cutting a cable in the Internet backbone could cause serious disruption, but since the Internet consists of interlinked networks, the rest of it would continue operating as usual.